

#14Days14Chapters

DAY 13

13 Feb } AD 14c Marathon complete
14 Feb }

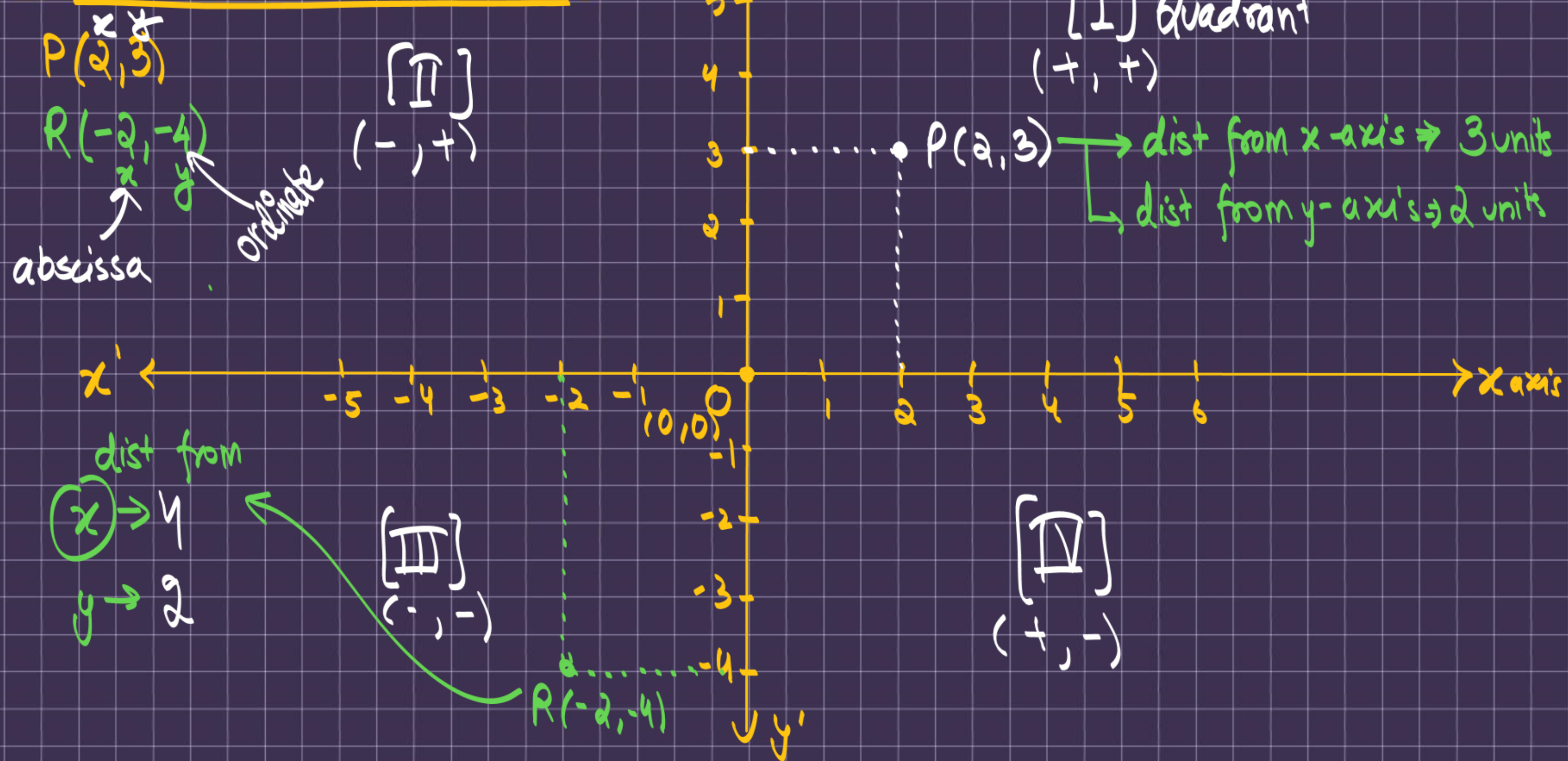
15 Feb → Maha Marathons.
16 Feb

17 Feb → 4AM

[Aim: 100/100 in Maths]

COORDINATE GEOMETRY

#Cartesian Plane:



LP:

$$S \overset{x}{(-3}, \overset{y}{4})$$

→ abscissa = x -coord $\Rightarrow -3$

→ ordinate = y -coord $\Rightarrow 4$

→ dist from y -axis = 3 units

→ " " x -axis = 4 units

→ Quadrant = Ind quad.

#(K3B)

x-axis eqⁿ $\rightarrow$ $y=0$

④ Any random point on x-axis $(x, 0)$

$x=0$ $\leftarrow$ eqⁿ of y-axis

$(0, 3)$

$(0, 2)$

$(0, 1)$

$(1, 0)$

$(3, 0)$

$(5, 0)$

eqⁿ of x-axis

$y=0$

⑤ y-axis eqⁿ $\Rightarrow$ $x=0$

any random point on y-axis $\Rightarrow (0, y)$

Coordinate Geo

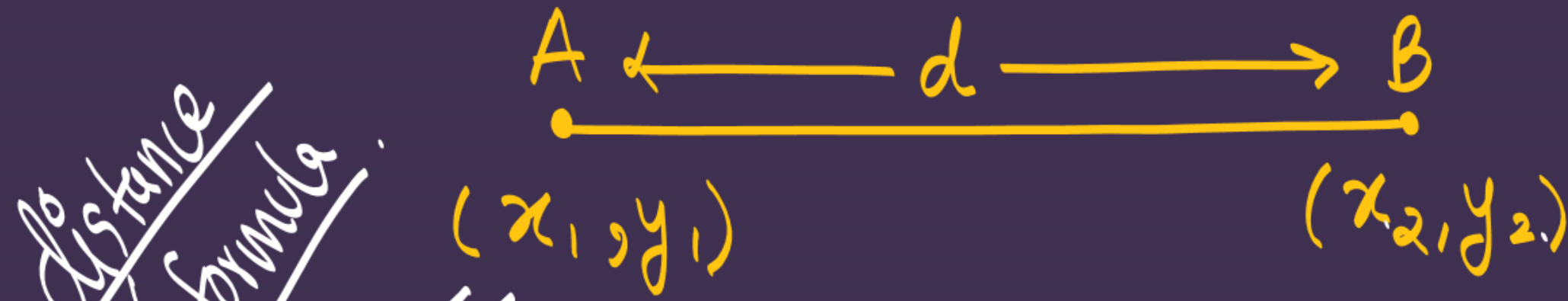
~~Distance formula~~

Section formula



$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

#Distance Between two points:



Distance
formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{or } \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$(9 - 2)^2 \Rightarrow 2^2 = \textcircled{4}$$

$$(2 - 4)^2 \Rightarrow (-2)^2 = \textcircled{4}$$

#LP: Find the distance between the points P(-4, 7) and Q(2, -5).



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(2 - (-4))^2 + (-5 - 7)^2}$$

$$d = \sqrt{(6)^2 + (-12)^2}$$

$$d = \sqrt{36 + 144}$$

$$d = \sqrt{180} \text{ units}$$
$$= 6\sqrt{5} \text{ units}$$

$$\sqrt{180} = \sqrt{2 \times 2 \times 3 \times 3 \times 5}$$

$$= \sqrt{2^2 \times 3^2 \times 5}$$

$$= 2 \times 3 \sqrt{5}$$

$$= 6\sqrt{5}$$

$$\begin{array}{r|l} 2 & 180 \\ \hline 2 & 90 \\ 3 & 45 \\ 3 & 15 \\ 5 & 3 \\ & 1 \end{array}$$

#LP: 1. The distance of the point $(-1, 7)$ from x-axis is

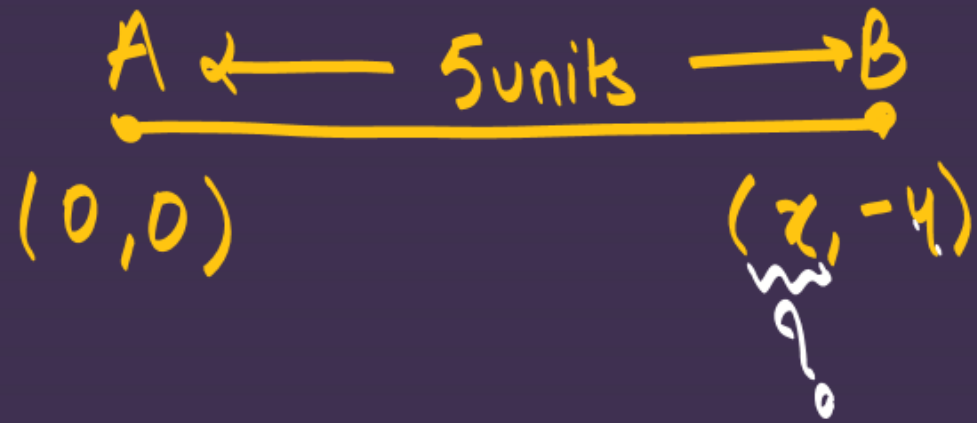
(a) -1

☒ (b) 7

(c) 6

(d) $\sqrt{50}$ (2023)

#LP: Find the value(s) of x, if the distance between the points A(0, 0) and B(x, -4) is 5 units. (2019)



$$d = 5$$

$$\Rightarrow \sqrt{(x-0)^2 + (-4-0)^2} = 5$$

$$\Rightarrow x^2 + (-4)^2 = (5)^2$$

$$\Rightarrow x^2 + 16 = 25$$

$$x^2 = 25 - 16$$

$$x^2 = 9$$

$$\boxed{x = \pm 3}$$

#LP:

The distance of a point A from x-axis is 3 units. Which of the following cannot be coordinates of the point A?

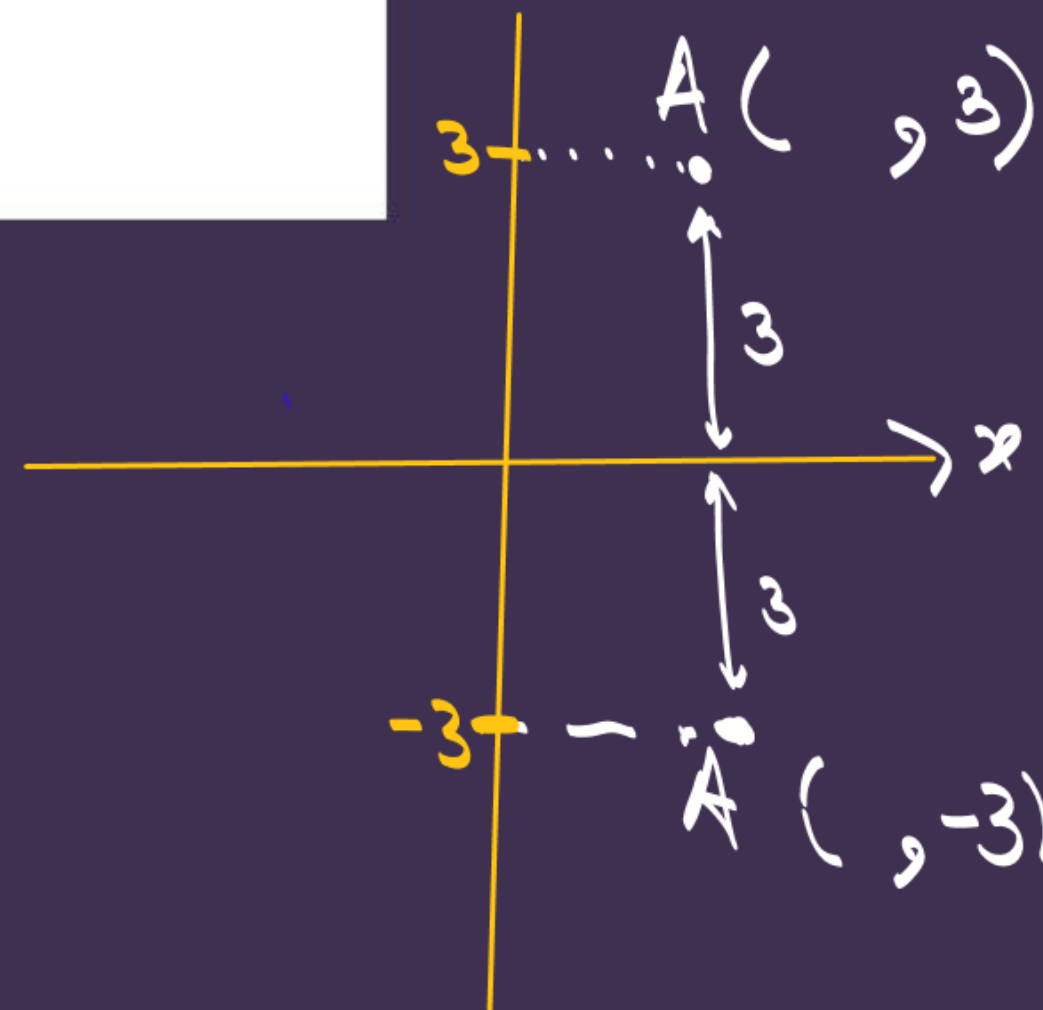
(a) $(1, 3)$

(b) $(-3, -3)$

(c) $(-3, 3)$

☒ (d) $(3, 1)$

dist from x-axis = 3 units



#LP: The centre of a circle is $(2a, a - 7)$. Find the values of ' a ' if the circle passes through the point $(11, -9)$.

Radius of the circle is $5\sqrt{2}$ cm. (3 Marks) (2023)

$$CP = 5\sqrt{2} \text{ cm}$$

$$\sqrt{(2a-11)^2 + (a-7 - (-9))^2} = 5\sqrt{2}$$

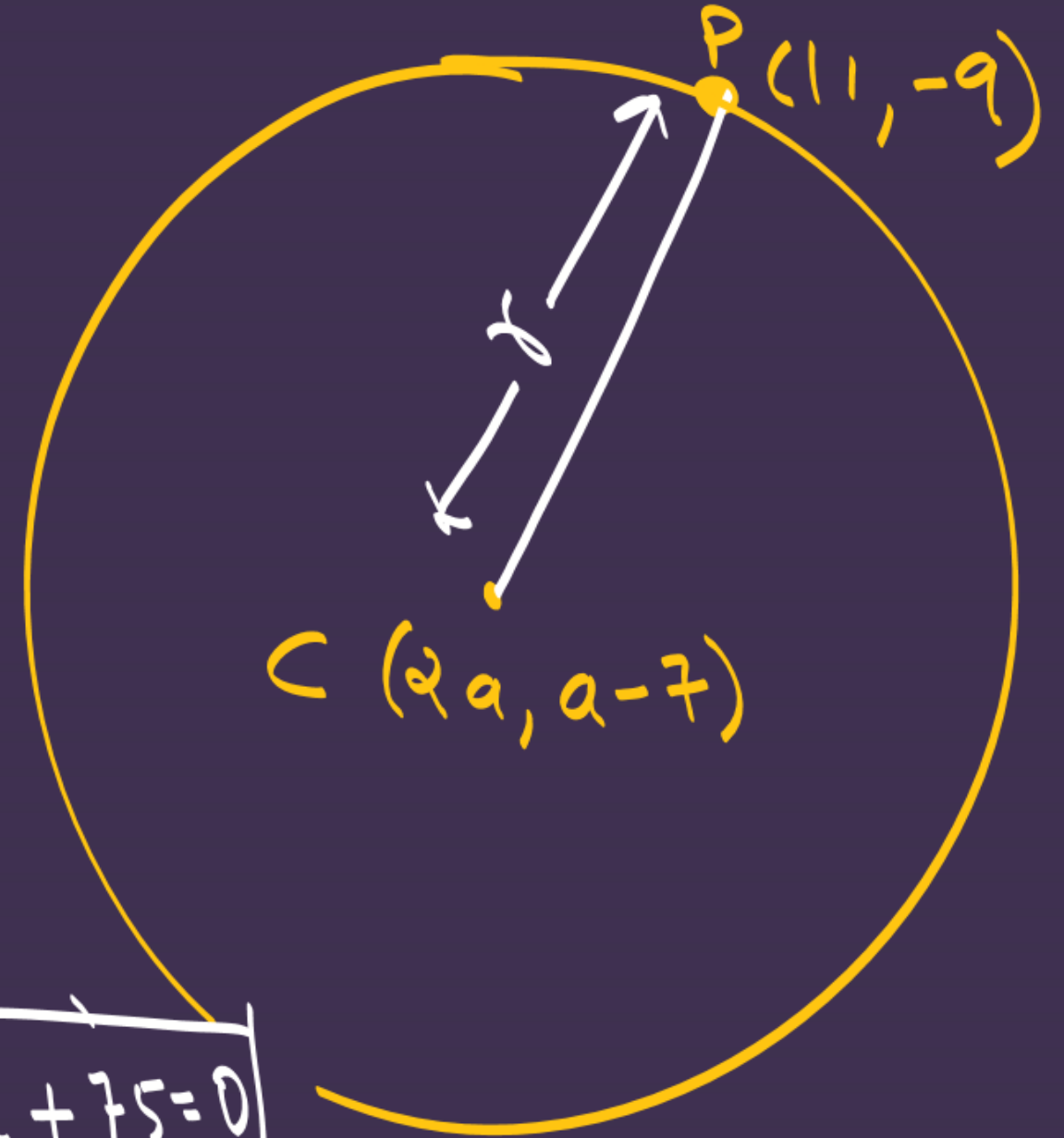
$$(2a-11)^2 + (a-7+9)^2 = (5\sqrt{2})^2$$

$$\Rightarrow (2a-11)^2 + (a+2)^2 = 25 \times 2$$

$$\Rightarrow (2a)^2 + (11)^2 - 2(2a)(11) + a^2 + (2)^2 + 2(a)(2) = 50$$

$$\Rightarrow \frac{4a^2 + 121 - 44a + a^2 + 4 + 4a}{2} = 50$$

$$\Rightarrow 5a^2 - 40a + 125 = 50 \Rightarrow 5a^2 - 40a + 75 = 0$$



$$5a^2 - 40a + 75 = 0$$
$$5(a^2 - 8a + 15) = 0$$

$$\boxed{a^2 - 8a + 15 = 0}$$

$$\underline{a^2 - 3a} - \underline{5a + 15} = 0$$

$$\underline{a(a-3)} - \underline{5(a-3)} = 0$$

$$(a-3)(a-5) = 0$$

$$\begin{array}{c|c} a-3=0 & a-5=0 \\ \hline a=3 & a=5 \end{array}$$

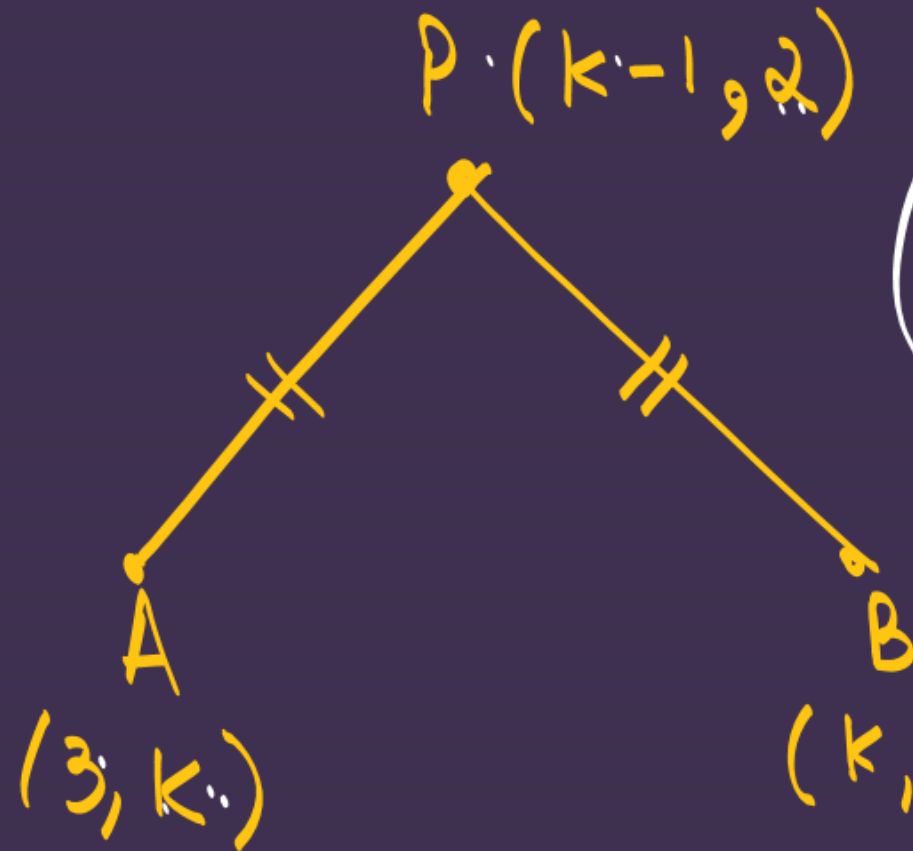
$$p+q = -8$$

$$pq = 15$$

$$\begin{array}{r|l} 3 & 15 \\ 5 & 15 \\ \hline & 1 \end{array}$$

#LP: If the point $P(k-1, 2)$ is equidistant from the points $A(3, k)$ and $B(k, 5)$, find the values of k .

[CBSE 2014]



$$PA = PB$$

$$\left(\sqrt{(k-1-3)^2 + (2-k)^2} \right) = \left(\sqrt{(k-1-k)^2 + (2-5)^2} \right)$$

squaring both sides

$$\Rightarrow (k-4)^2 + (k-2)^2 = (-1)^2 + (-3)^2$$

$$\Rightarrow k^2 + 16 - 8k + k^2 + 4 - 4k = 1 + 9$$

$$\Rightarrow 2k^2 - 12k + 20 = 10$$

$$\Rightarrow 2k^2 - 12k + 10 = 0$$

$$\Rightarrow 2(k^2 - 6k + 5) = 0$$

$$k^2 - 6k + 5 = 0$$

$$k^2 - 5k - k + 5 = 0$$

$$k(k-5) - 1(k-5) = 0$$

$$(k-5)(k-1) = 0$$

$$\begin{array}{l} k-5=0 \\ \hline k=5 \end{array}$$

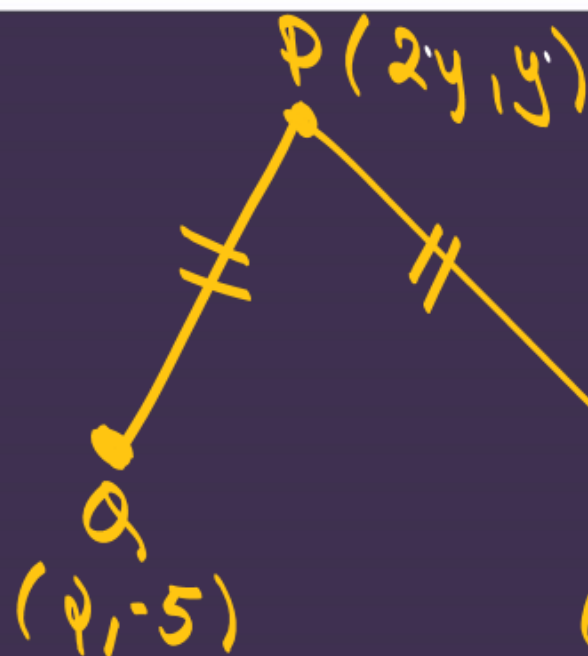
$$\begin{array}{l} k-1=0 \\ \hline k=1 \end{array}$$

#LP: Q3: The x-coordinate of a point P is twice its y-coordinate. If P is equidistant from Q(2, -5) and R(-3, 6), then the coordinates of P are (2022)

- (a) (8, 16)
- (b) (10, 20)
- (c) (20, 10)
- ☒ (d) (16, 8)

$$P(2y, y)$$

$$P(2y, y) \\ (2(8), 8) \\ P(16, 8)$$



$$PQ = PR$$

squaring both sides

$$PQ^2 = PR^2$$

$$(2y-2)^2 + (y+5)^2 = (2y+3)^2 + (y-6)^2$$

$$\Rightarrow \cancel{4y^2} + 4 - 8y + \cancel{y^2} + 25 + 10y = \cancel{4y^2} + 9 + 12y + \cancel{y^2} + 36 - 12y$$

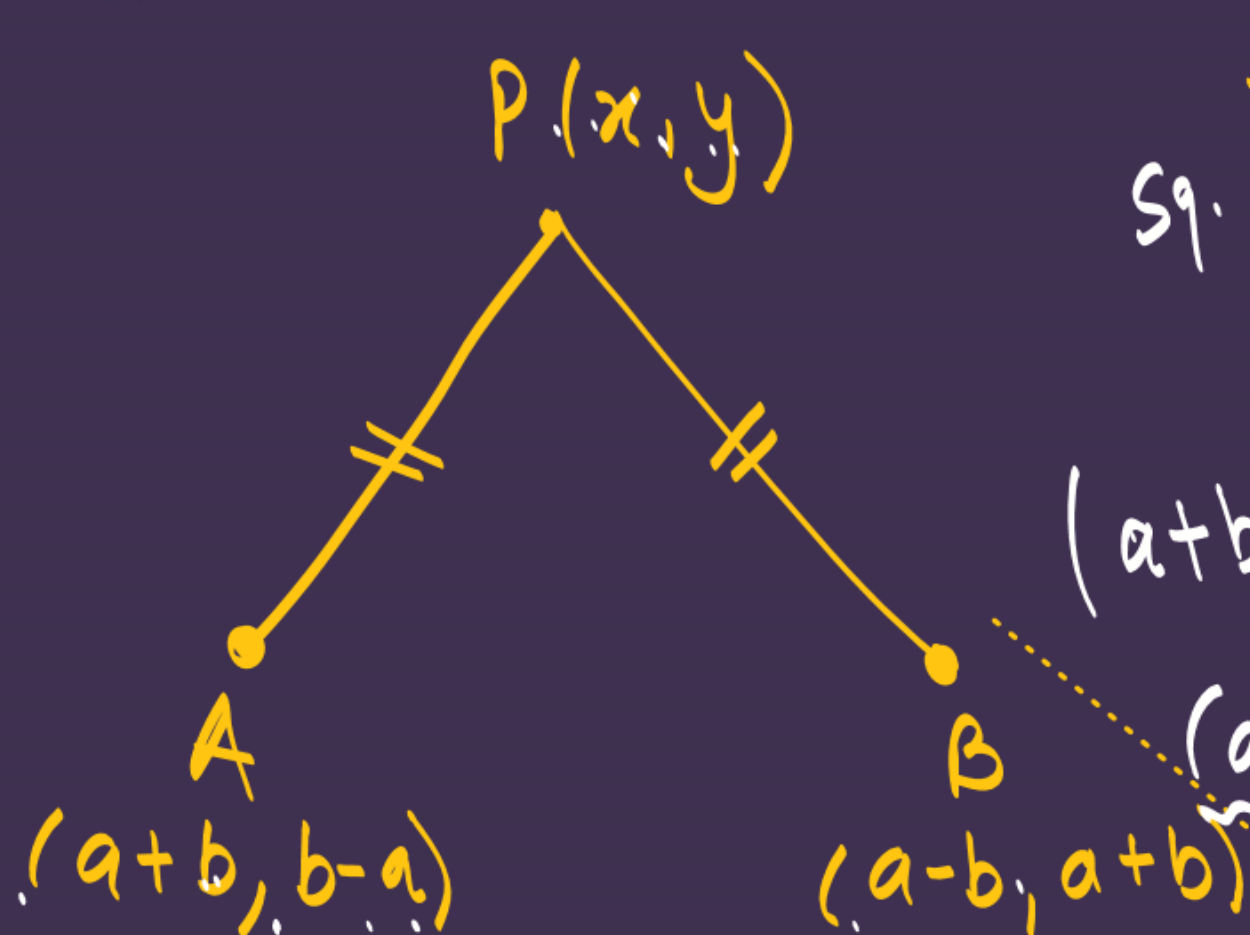
$$2y + 29 = 45$$

$$2y = 45 - 29$$

$$2y = 16$$

$$y = 8$$

#LP: If the point $P(x, y)$ is equidistant from the points $A(a+b, b-a)$ and $B(a-b, a+b)$, then prove that $bx=ay$.



$$PA = PB$$

Sq. both sides

$$PA^2 = PB^2$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$(a+b-x)^2 + (b-a-y)^2 = (a-b-x)^2 + (a+b-y)^2$$

$$(a+b-x)^2 - (a-b-x)^2 = (a+b-y)^2 - (b-a-y)^2$$

$$A^2 - B^2 \Rightarrow (A+B)(A-B)$$

$$\Rightarrow (a+b-x+a-b-x)(a+b-x-a-b+x) = (a+b-y+b-a-y)$$

$$\Rightarrow (2a-2x)(\cancel{a+b-x} - \cancel{a+b-x}) = (a+b-y-b-a-y)$$

$$\Rightarrow \cancel{2}(a-x)(\cancel{b}) = \cancel{2}(b-y)(\cancel{a})$$

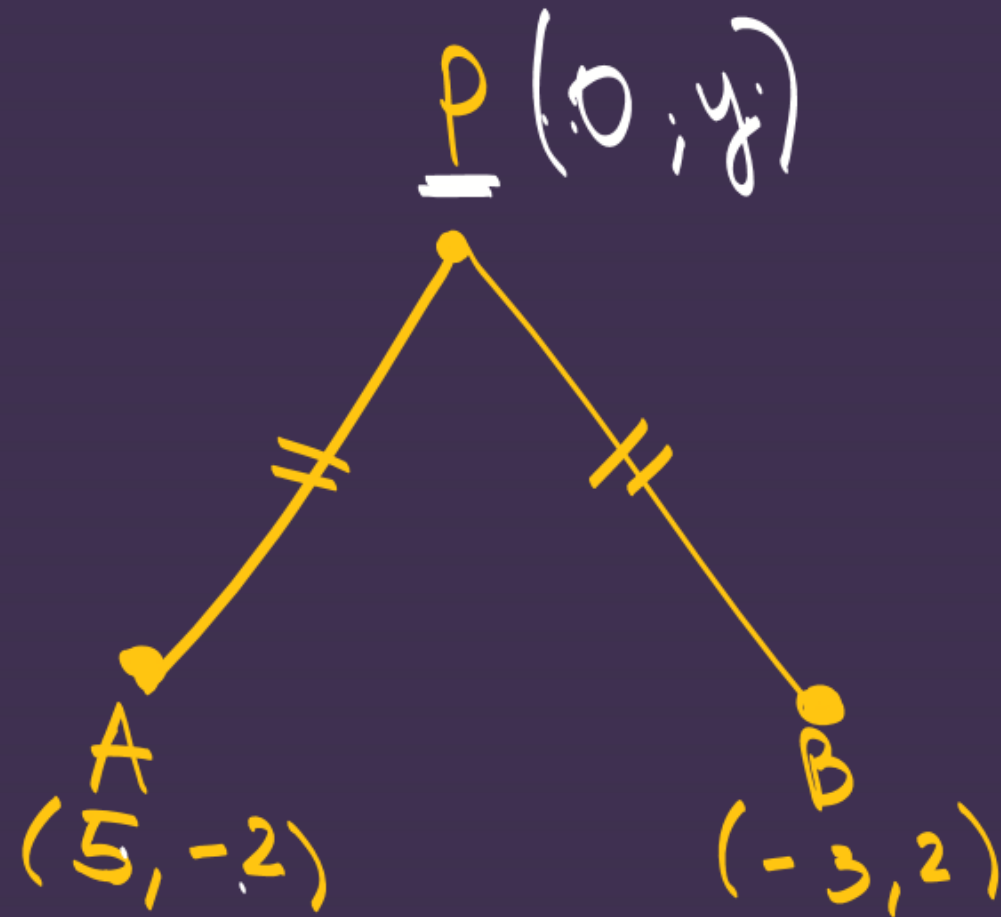
$$(a-x)b = (b-y)a$$

$$\cancel{ab} - bx = \cancel{ab} - ay$$

$$-bx = -ay$$

$$bx = ay \quad \text{Q.E.D.}$$

#LP: Find the point on y-axis which is equidistant from the points (5, -2) and (-3, 2). (Delhi 2019)



$$\text{Let } P(0, y)$$

$$\text{given } PA = PB$$

sq. both sides

$$PA^2 = PB^2$$

$$(5-0)^2 + (y-(-2))^2 = (-3-0)^2 + (y-2)^2$$

$$25 + (y+2)^2 = 9 + (y-2)^2$$

$$25 + \cancel{y^2} + \cancel{4y} + 4 = 9 + \cancel{y^2} + \cancel{4y} - 4$$

$$25 - 9 = -4y - 4y$$

$$16 = -8y$$

$$\boxed{-2 = y}$$

Final Ans

$$\boxed{P(0, -2)}$$

#LP:

Find the point on x-axis which is equidistant from point A(2,-2) and B(-4,2).

let $P(x, 0)$

[CBSE 2020-21]

#LP: The distance between the points $(0, 0)$ and $(a - b, a + b)$ is (2020)

(a) $2\sqrt{ab}$

(b) $\sqrt{2a^2 + ab}$

(c) $2\sqrt{a^2 + b^2}$

~~(d)~~ $\sqrt{2a^2 + 2b^2}$

$$AB = \sqrt{(a-b-0)^2 + (a+b-0)^2}$$

$$AB = \sqrt{(a-b)^2 + (a+b)^2}$$

$$AB = \sqrt{\underline{a^2} + \underline{b^2} - \cancel{2ab} + \underline{a^2} + \underline{b^2} + \cancel{2ab}}$$

$$AB = \sqrt{2a^2 + 2b^2}$$

$$= \sqrt{2(a^2 + b^2)}$$

$$= \sqrt{2} \sqrt{a^2 + b^2}$$

$$\Rightarrow \frac{\sqrt{2a}}{2\sqrt{a}} \quad (A)$$

$$\frac{\sqrt{2^2 a}}{2\sqrt{a}} \quad (B)$$

#LP: The distance between the points $(\underbrace{a \cos \theta + b \sin \theta}_A, 0)$ and $(0, \underbrace{a \sin \theta - b \cos \theta}_B)$, is

(A) $a^2 + b^2$

(B) $a^2 - b^2$

☒ (C) $\sqrt{a^2 + b^2}$

(D) $\sqrt{a^2 - b^2}$

$$AB = \sqrt{(a \cos \theta + b \sin \theta - 0)^2 + (a \sin \theta - b \cos \theta - 0)^2}$$

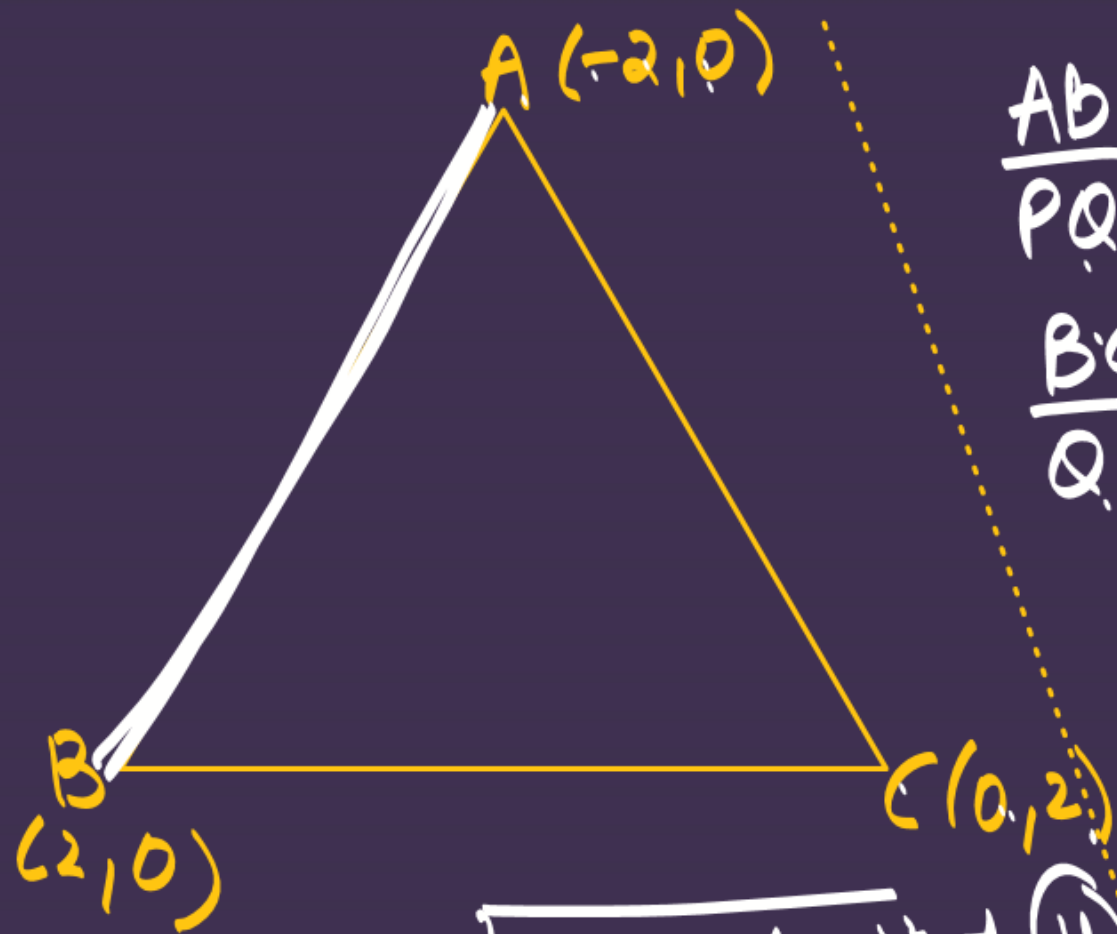
$$= \sqrt{(a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2}$$

$$= \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta + \cancel{2a \cos \theta b \sin \theta} + a^2 \sin^2 \theta + b^2 \cos^2 \theta - \cancel{2a \sin \theta b \cos \theta}}$$

$$= \sqrt{a^2 (\underbrace{\cos^2 \theta + \sin^2 \theta}_{=1}) + b^2 (\underbrace{\sin^2 \theta + \cos^2 \theta}_{=1})}$$

$$AB \Rightarrow \sqrt{a^2 + b^2}$$

#LP: Show that $\triangle ABC$, where $A(-2, 0)$, $B(2, 0)$, $C(0, 2)$ and $\triangle PQR$ where $P(-4, 0)$, $Q(4, 0)$, $R(0, 4)$ are similar triangles. (CBSE 2017)



$$AB = \sqrt{(-2-2)^2 + (0-0)^2} \Rightarrow 4$$

$$BC = \sqrt{(0-2)^2 + (2-0)^2} \Rightarrow \sqrt{8}$$

$$CA = \sqrt{(-2-0)^2 + (0-2)^2} \Rightarrow \sqrt{8}$$

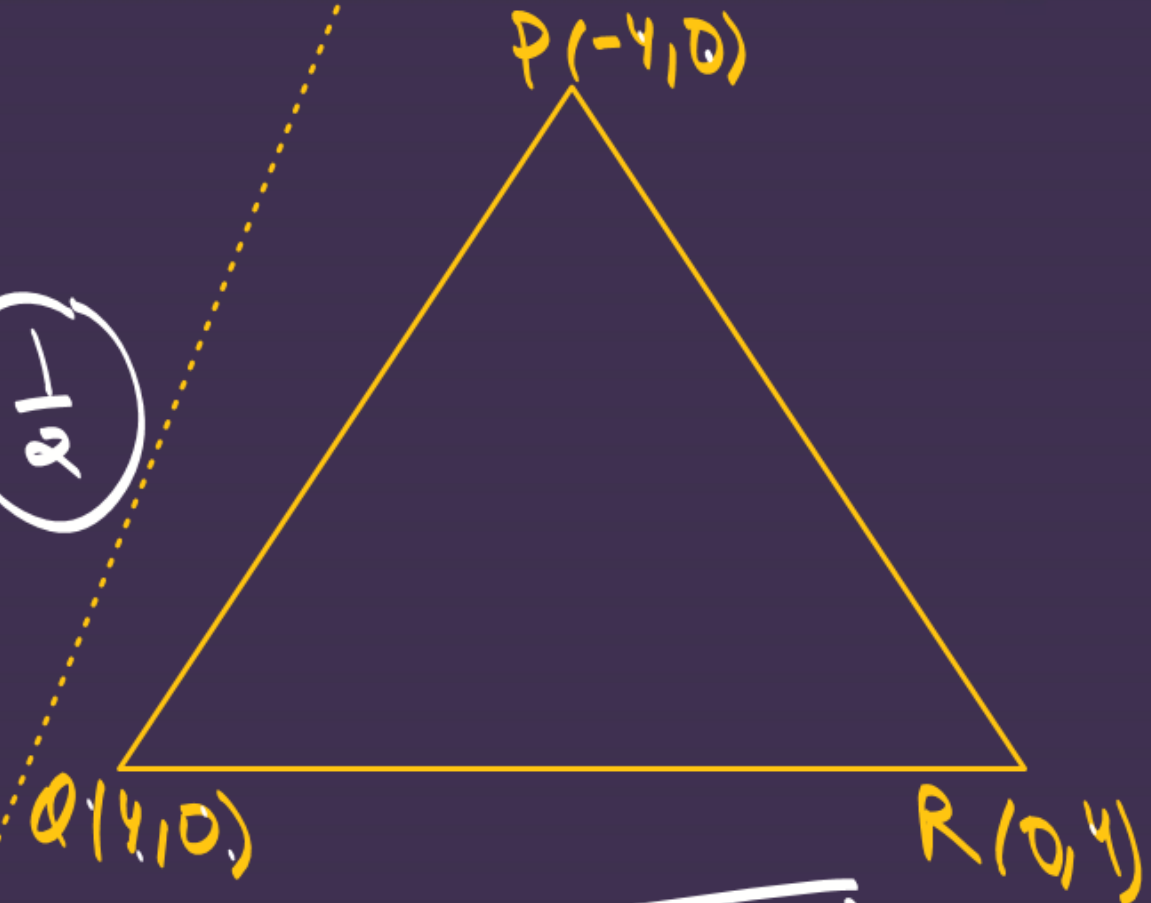
$$\frac{AB}{PQ} = \frac{4}{8} \Rightarrow \frac{1}{2}$$

$$\frac{BC}{QR} = \frac{\sqrt{8}}{\sqrt{32}} \Rightarrow \sqrt{\frac{8}{32}} \Rightarrow \sqrt{\frac{1}{4}} \Rightarrow \frac{1}{2}$$

$$\frac{AC}{PR} = \frac{\sqrt{8}}{\sqrt{32}} = \frac{1}{2}$$

$$\therefore \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

$\therefore$ SSS $\Rightarrow \triangle ABC \sim \triangle PQR$



$$PQ = \sqrt{(-4-4)^2 + (0-0)^2} \Rightarrow 8$$

$$\frac{QR}{PR} = \frac{\sqrt{(4-0)^2 + (0-4)^2}}{\sqrt{32}}$$

#Figures ki Jaankari: (for this ^{only} chapter)

//gm → both pair of opp sides ki length equal

Rect → opp sides equal + diagonals equal.

Square → all sides equal + diagonals equal.

Rhombus → all sides equal.

Right Δ → Pythag. $H^2 = P^2 + B^2$

Isosc. Δ → any two sides length eq

Eq. Δ → all 3 sides equal

#LP: Show that the points ^A(7, 10), ^B(-2, 5) and ^C(3, -4) are vertices of an isosceles right triangle. (2020)

$$AB = \sqrt{(7+2)^2 + (10-5)^2}$$
$$= \sqrt{9^2 + 5^2} \Rightarrow \sqrt{81+25} \Rightarrow \sqrt{106}$$

$$BC = \sqrt{(-2-3)^2 + (5+4)^2}$$
$$\Rightarrow \sqrt{(-5)^2 + (9)^2} \Rightarrow \sqrt{25+81} \Rightarrow \sqrt{106}$$

$$AC = \sqrt{(7-3)^2 + (10+4)^2}$$
$$= \sqrt{4^2 + 14^2}$$
$$\Rightarrow \sqrt{16+196} \Rightarrow \sqrt{212}$$

any two
sides eq.

AB=BC
∴ Isos. Δ ✓
(HP)

Pythm.

$$H^2 = P^2 + B^2$$

we can clearly see

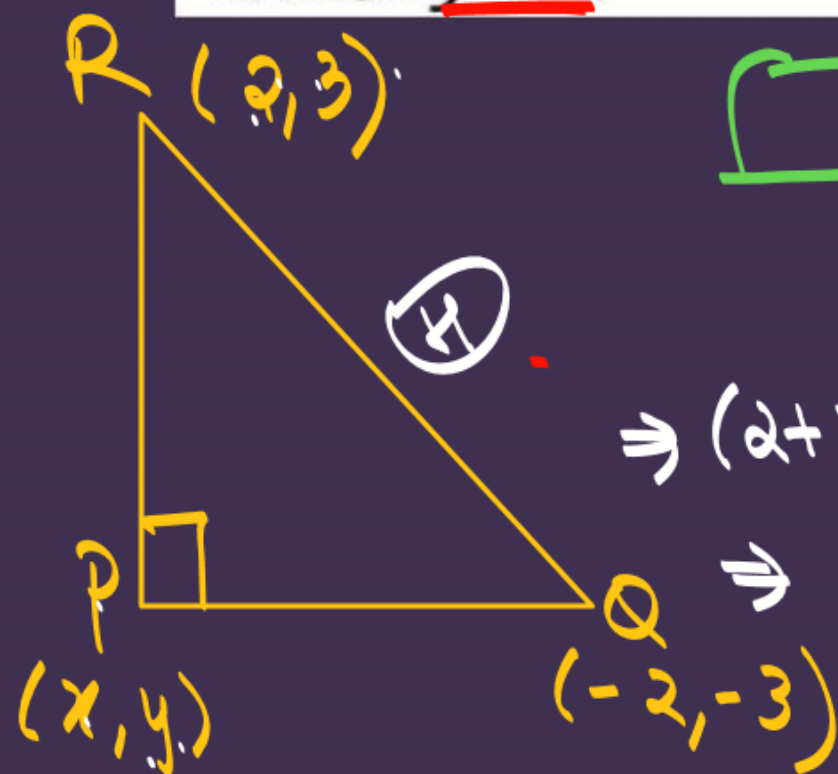
$$\boxed{AC^2 = AB^2 + BC^2}$$
$$(\sqrt{212})^2 = (\sqrt{106})^2 + (\sqrt{106})^2$$

$$212 = 106 + 106$$

$$212 = 212 \quad \text{(HP)}$$

∴ it's right Δ.

#LP: P (x, y), Q (-2, -3) and R (2, 3) are the vertices of a right triangle PQR right angled at P. Find the relationship between x and y. Hence, find all possible values of x for which y = 2.



Pytha $\rightarrow H^2 = P^2 + B^2$

$$(RQ)^2 = PR^2 + PQ^2$$

$$\Rightarrow (2+2)^2 + (3+3)^2 = [(x-2)^2 + (y+3)^2] + [(x+2)^2 + (y+3)^2]$$

$$\Rightarrow 4^2 + 6^2 = \cancel{x^2} + \cancel{4} - \cancel{4x} + \cancel{y^2} + \cancel{9} - \cancel{6y} + \cancel{x^2} + \cancel{4} + \cancel{4x} + \cancel{y^2} + \cancel{9} + \cancel{6y}$$

$$\Rightarrow 16 + 36 = 2x^2 + 2y^2 + 8 + 18$$

$$\Rightarrow 52 = 2(x^2 + y^2) + 26$$

$$\Rightarrow 52 - 26 = 2(x^2 + y^2)$$

$$\Rightarrow 13 = x^2 + y^2$$

$$x^2 + y^2 = 13$$

put $y = 2$

$$x^2 + 2^2 = 13$$

$$x^2 + 4 = 13$$

$$x^2 = 13 - 4$$

$$x^2 = 9$$

$$x = \pm 3$$

#LP:

If $(-5, 3)$ and $(5, 3)$ are two vertices of an equilateral triangle, then find coordinates of the third vertex, given that origin lies inside the triangle
(Take $\sqrt{3} = 1.7$)

[CBSE 2014]

Q $(-5, 3)$ R $(5, 3)$ Eq. $\Delta \Rightarrow PQ = QR = PR$
 $PQ^2 = QR^2 = PR^2$
 $(x+5)^2 + (y-3)^2 = (5-(-5))^2 + (3-3)^2 = (x-5)^2 + (y-3)^2$
 $x^2 + 25 + 10x + y^2 + 9 - 6y = 100$ $= x^2 + 25 - 10x + y^2 + 9 - 6y$
 $x^2 + y^2 + 10x - 6y + 34 = 100$ $= x^2 + y^2 - 10x - 6y + 34$
 $\text{I} = \text{II}$
 $x^2 + y^2 + 10x - 6y + 34 = 100$ $\Rightarrow y^2 - 6y - 66 = 0$
 $\text{I} = \text{II}$
 $x^2 + y^2 + 10x - 6y + 34 = 100$ $\Rightarrow y^2 - 6y - 66 = 0$
 $0 + y^2 + 0 - 6y = 100 - 34$
 $y^2 - 6y = 66$
 $10x = -10x \Rightarrow 20x = 0$
 $x = 0$

$$y^2 - 6y - 66 = 0$$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$y = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(-66)}}{2(1)}$$

$$y = \frac{6 \pm \sqrt{36 + 264}}{2}$$

$$y = \frac{6 \pm \sqrt{300}}{2}$$

$$y = \frac{6 \pm 10\sqrt{3}}{2}$$

$$y = \frac{6 + 10\sqrt{3}}{2}$$

$$= \frac{2(3 + 5\sqrt{3})}{2}$$

$$y = \frac{6 - 10\sqrt{3}}{2}$$

$$= \frac{2(3 - 5\sqrt{3})}{2}$$

$$y = \underline{3 + 5\sqrt{3}}, \underline{3 - 5\sqrt{3}}$$

$$p + q = -6$$

$$pq = 66$$

$$(22) \begin{pmatrix} 2 & 66 \\ 3 & 33 \\ 11 & 11 \end{pmatrix}$$

$$\sqrt{300} = \sqrt{2^2 \cdot 3 \cdot 5^2}$$

$$= 2 \cdot 5 \sqrt{3}$$

$$\sqrt{300} = 10\sqrt{3}$$

$$\begin{pmatrix} 2 & 300 \\ 2 & 150 \\ 3 & 75 \\ 5 & 25 \\ 5 & 5 \\ 1 & 1 \end{pmatrix}$$

$$y = 3 + 5\sqrt{3}$$

$$y = 3 + 5(1.7)$$

$$y = 3 + 8.5$$

$$\boxed{y = 11.5}$$

$$y = 3 - 5\sqrt{3}$$

$$= 3 - 5(1.7)$$

$$= 3 - 8.5$$

$$\boxed{y = -5.5}$$

$$\therefore \underline{\underline{P(0, -5.5)}}$$

#LP: Show that the point ^A(1, -1), ^B(5, 2) and ^C(9, 5) are collinear. [CBSE 2006]

$$AB = \sqrt{(5-1)^2 + (2+1)^2} \\ = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$BC = \sqrt{(9-5)^2 + (5-2)^2} \\ = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$CA = \sqrt{(9-1)^2 + (5+1)^2} \\ = \sqrt{8^2 + 6^2} = \sqrt{64+36} = \sqrt{100} = 10$$

any two
sum = third

we can clearly see

$$AB + BC = AC$$

Hence, A, B, C are
collinear

#Section Formula:

$$\frac{AP}{PB} = \frac{m}{n}$$



Section
Formula →

$$P \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

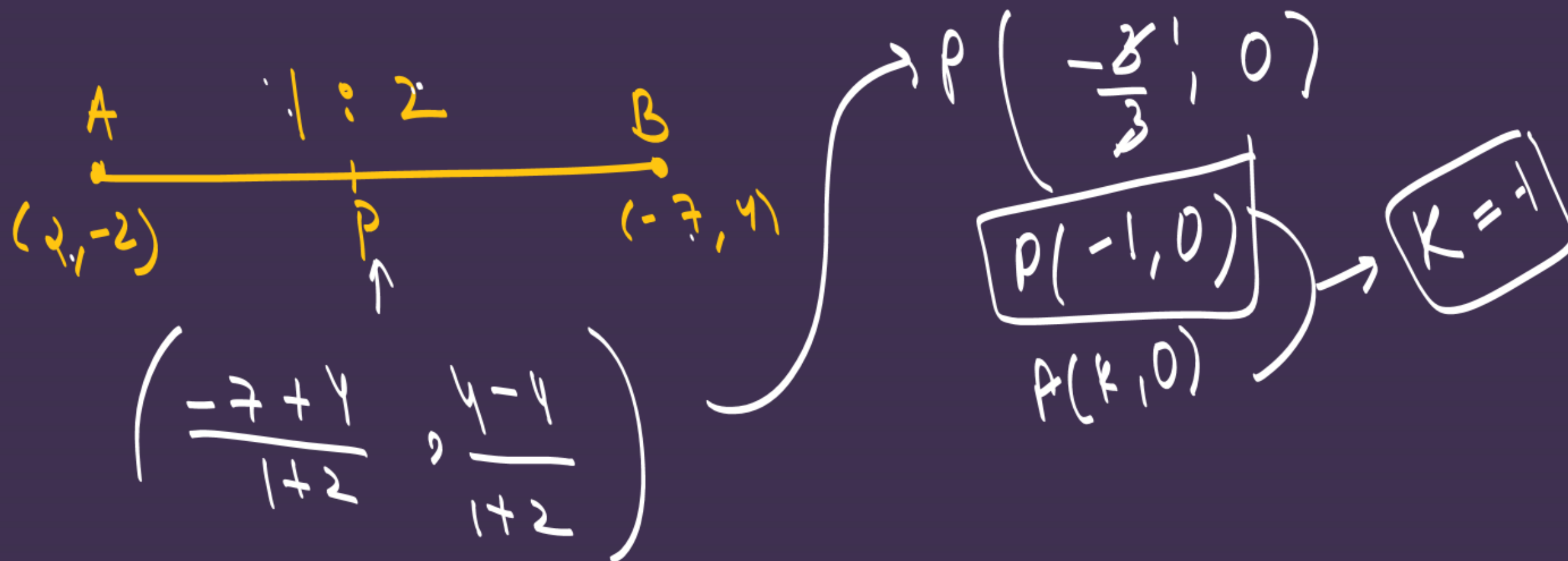
#LP: If the point $P(k, 0)$ divides the line segment joining the points $A(2, -2)$ and $B(-7, 4)$ in the ratio $1:2$ then the value of k is (2020)

(a) 1

(b) 2

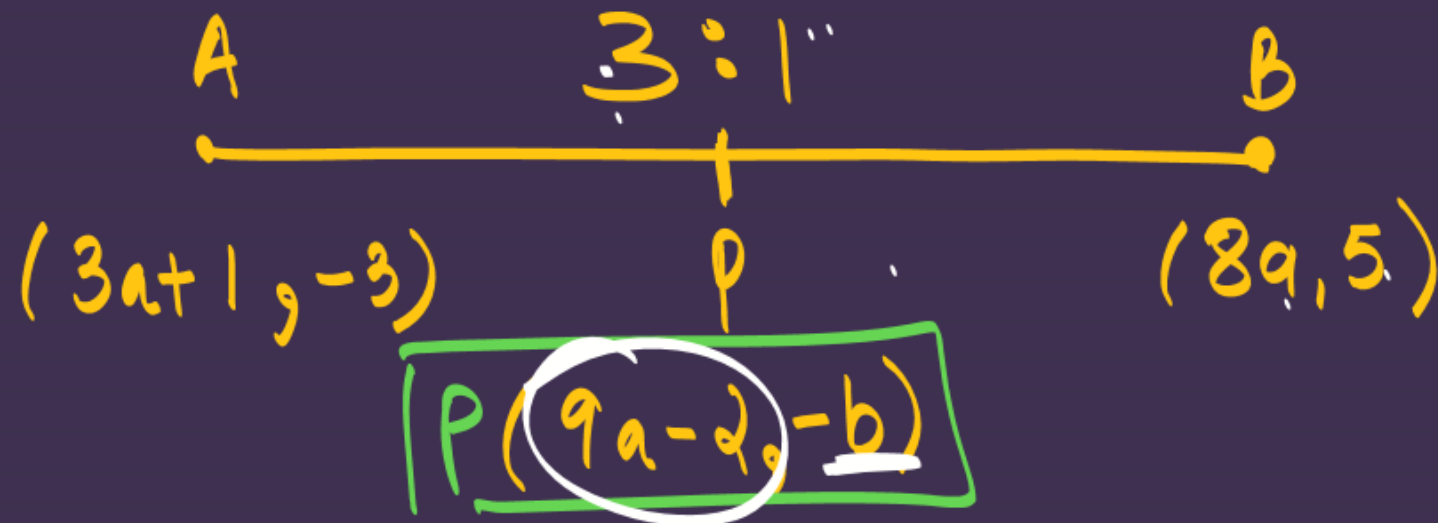
(c) -2

~~(d) -1~~



#LP: If $P(9a - 2, -b)$ divides line segment joining $A(3a + 1, -3)$ and $B(8a, 5)$ in the ratio $3 : 1$, find the values of a and b .

[CBSE 2016]



$$P \left(\frac{24a + 1(3a+1)}{3+1}, \frac{15 + (-3)}{3+1} \right)$$

Comparing x coord

$$9a - 2 = \frac{24a + 3a + 1}{4}$$

$$\Rightarrow 4(9a - 2) = 27a + 1$$

$$\Rightarrow 36a - 8 = 27a + 1$$

$$\Rightarrow 36a - 27a = 8 + 1$$

$$\Rightarrow 9a = 9$$

$$\Rightarrow a = 1$$

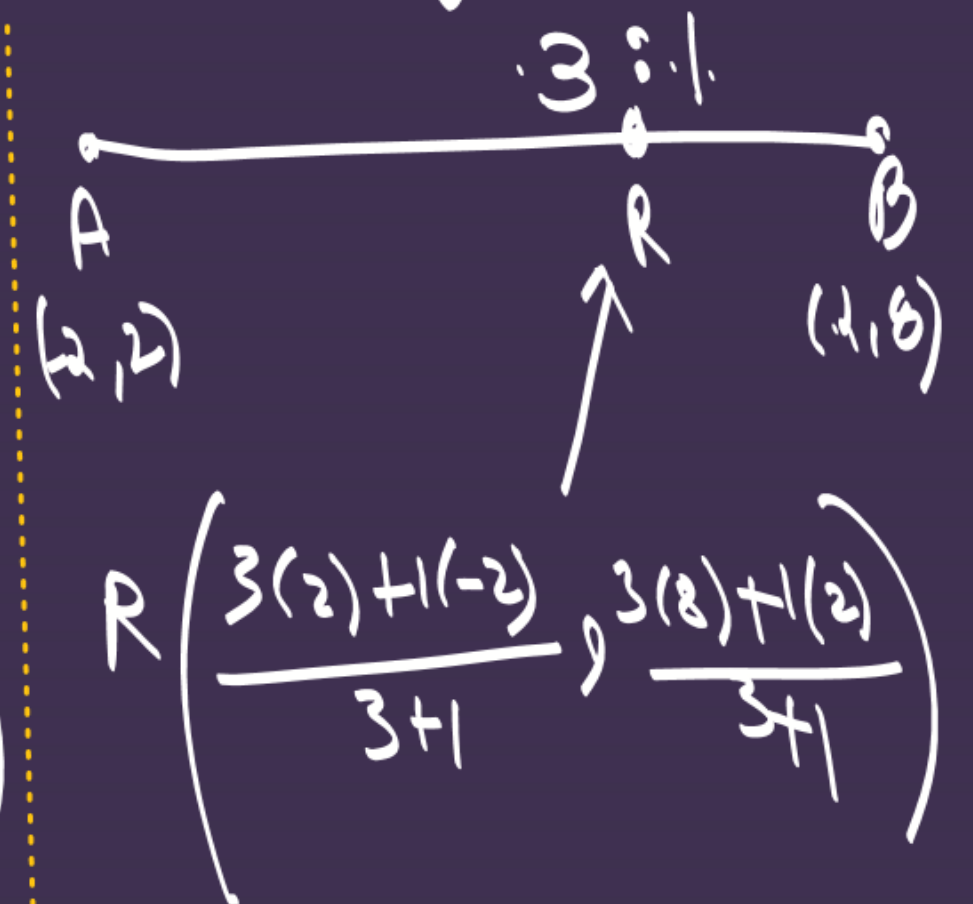
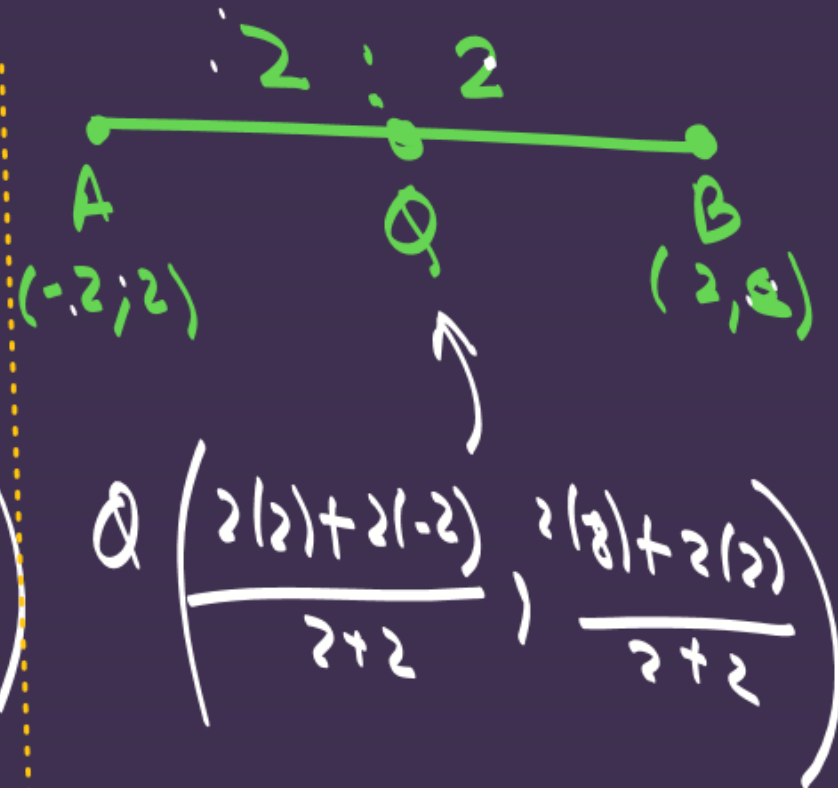
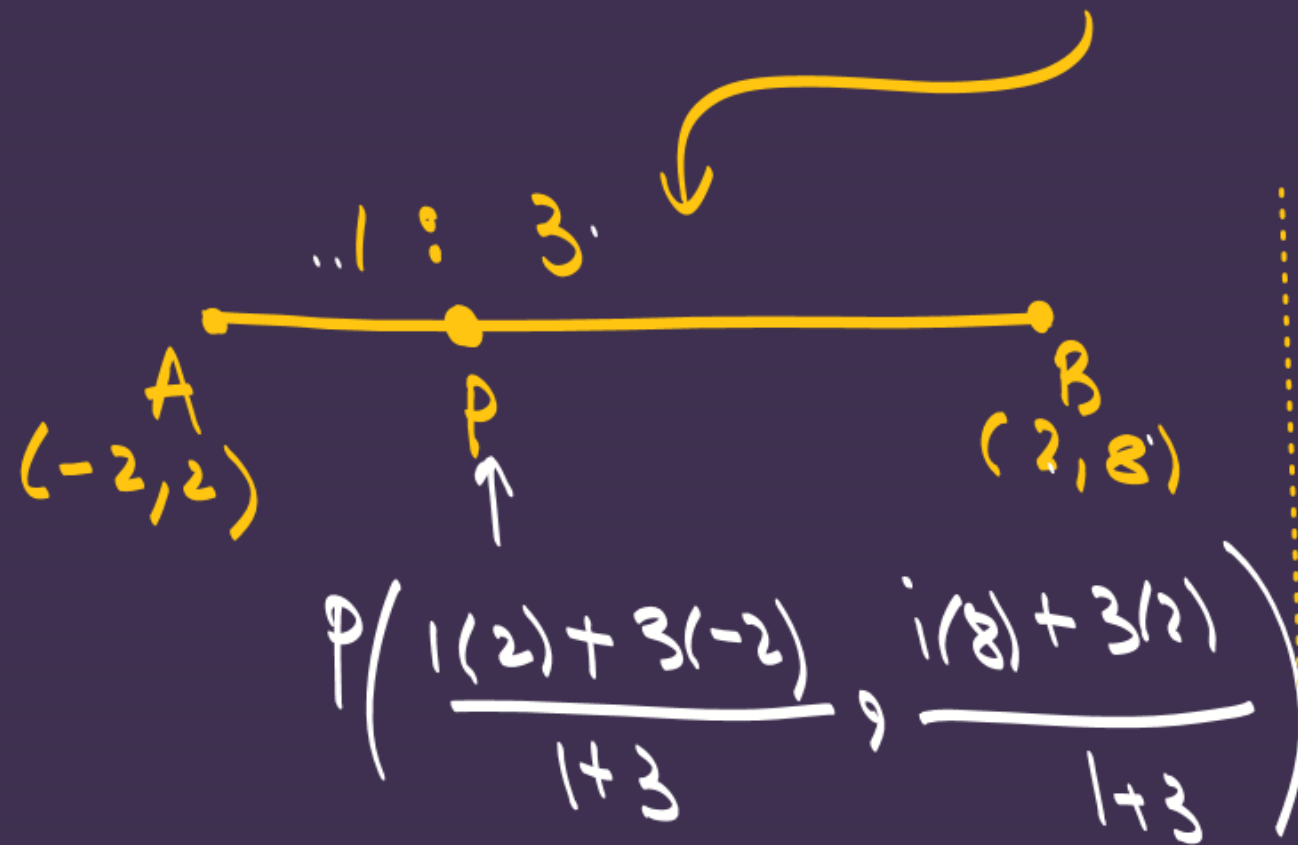
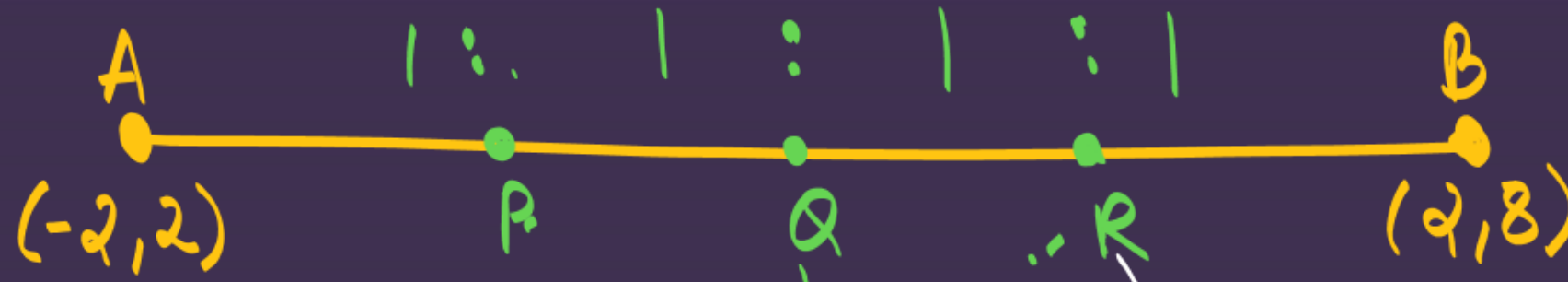
comparing y-word

$$-b = \cancel{1}^3$$

$$\boxed{b = -3}$$

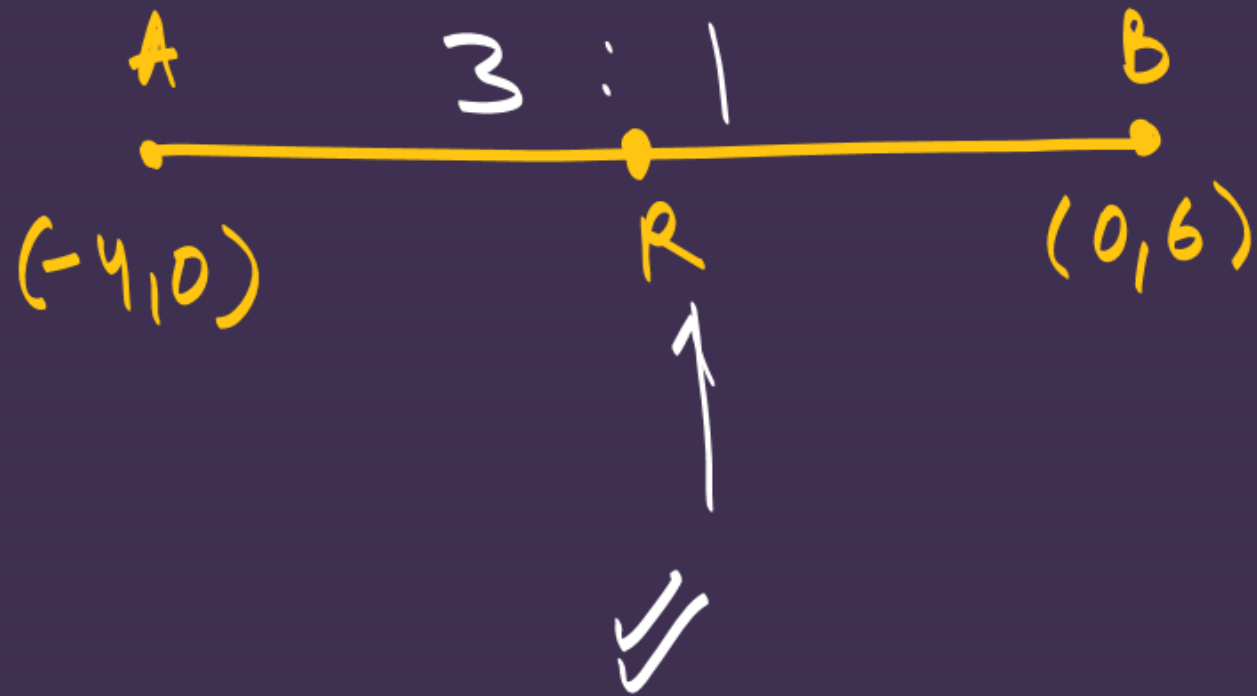
$$\left. \begin{array}{l} a = 1 \\ b = -3 \end{array} \right\} \text{done}$$

#LP: Find the coordinates of the points which divide the line segment joining A(-2, 2) and B(2, 8) into four equal parts.



#LP: The point R divides the line segment AB, where A (-4, 0) and B(0, 6) such that $AR = \frac{3}{4} AB$.

Find the coordinates of R. (CBSE 2019)



$$AR = \frac{3}{4} AB$$

$$\frac{AR}{AB} = \frac{3}{4}$$

$$\Rightarrow \frac{AR}{AR+RB} = \frac{3}{4}$$

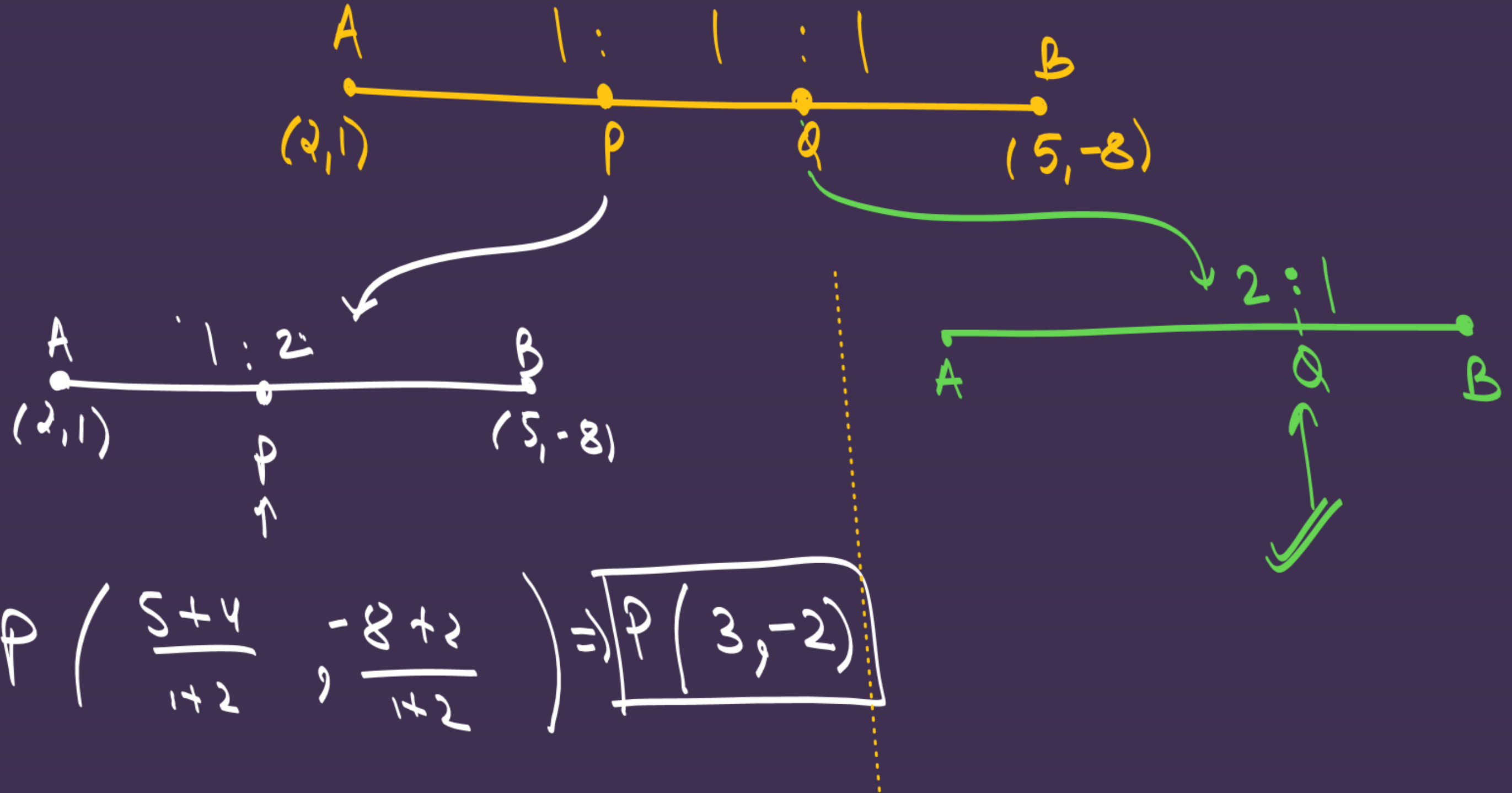
$$\rightarrow 4AR = 3AR + 3RB$$

$$\rightarrow 4AR - 3AR = 3RB$$

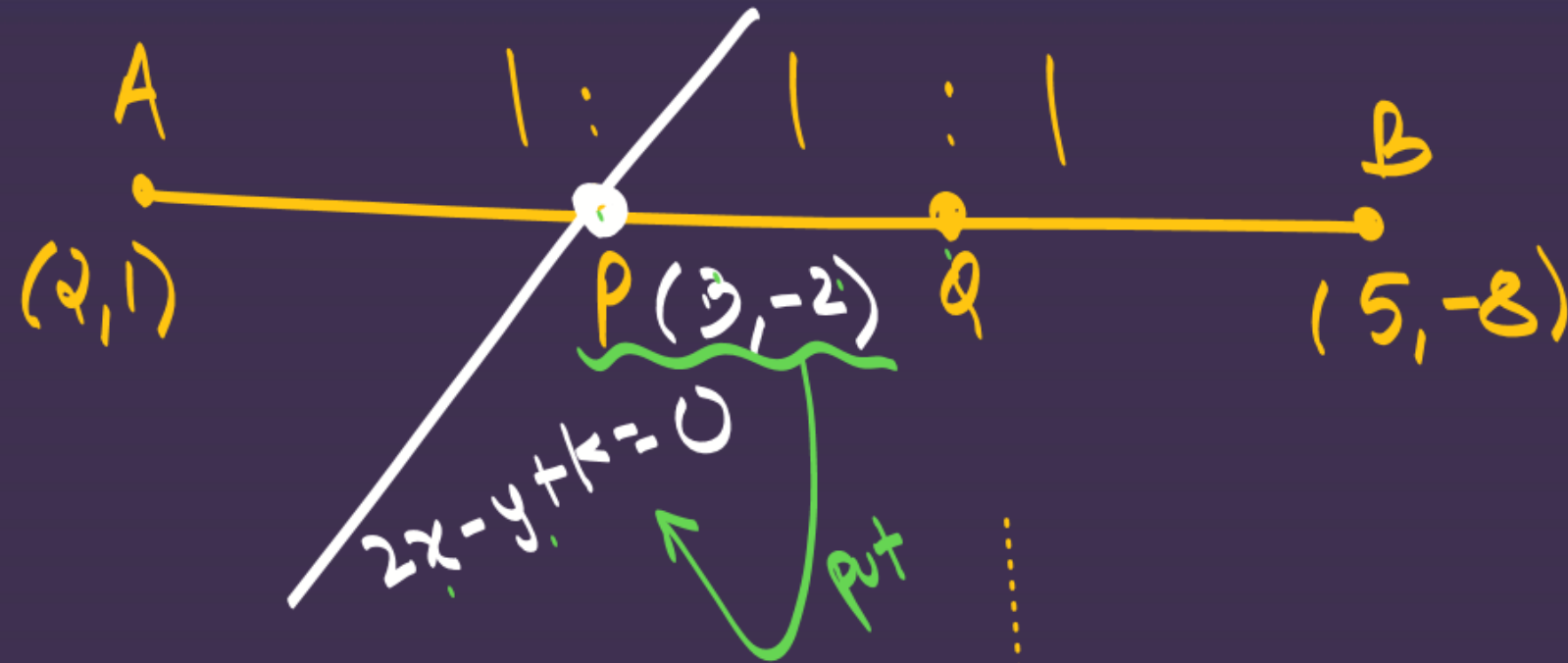
$$AR = 3RB$$

$$\frac{AR}{RB} = \frac{3}{1}$$

#LP: The line segment joining the points A(2, 1) and B(5, -8) is trisected at the points P and Q such that P is nearer to A. (2019)



#LP: The line segment joining the points $A(2, 1)$ and $B(5, -8)$ is trisected at the points P and Q such that P is nearer to A . If P also lies on the line given by $2x - y + k = 0$, find the value of k . (2019)



$$2(3) - (-2) + k = 0$$

$$6 + 2 + k = 0$$

$$8 + k = 0$$

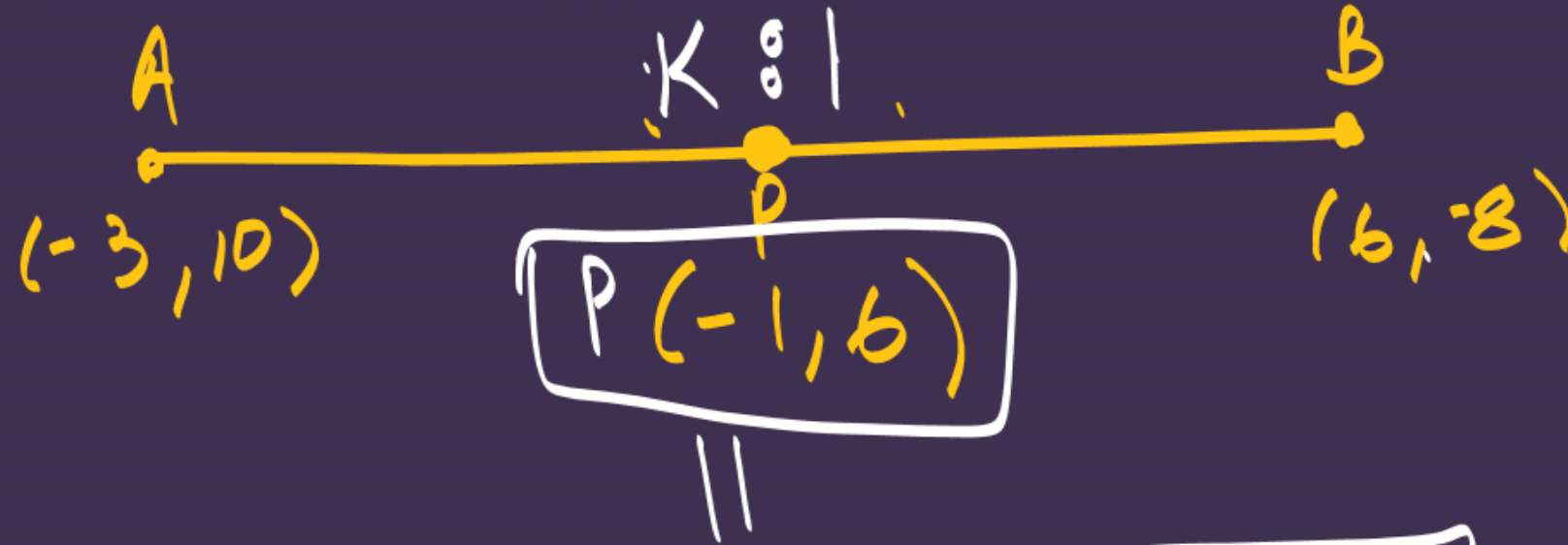
$$\boxed{k = -8}$$

#Finding Ratio:

#LP: Find the ratio in which the line segment joining the points $A(-3, 10)$ and $B(6, -8)$ is divided by $P(-1, 6)$.

let

$$\frac{AP}{PB} = \frac{k}{1}$$



$$P \left(\frac{6k-3}{k+1}, \frac{-8k+10}{k+1} \right)$$

comparing 'x'

$$\frac{6k-3}{k+1} = -1$$

$$6k-3 = -k-1$$

$$6k+k = 3-1$$

$$7k = 2$$

$$k = \frac{2}{7}$$

$$\boxed{2:7}$$

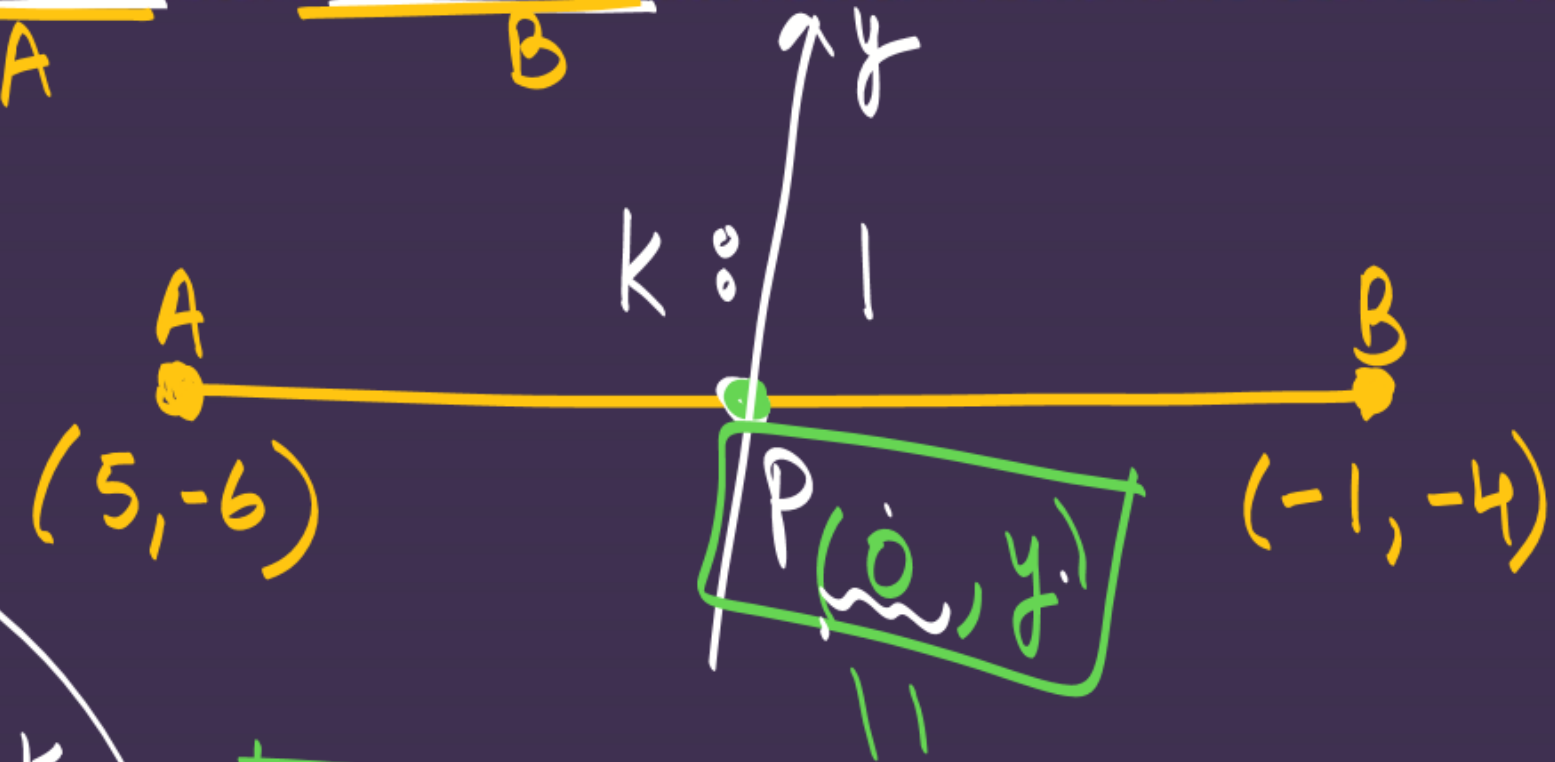
$$\frac{AP}{PB} = \frac{K}{1}$$

$$\frac{AP}{PB} = \frac{2/7}{1}$$

$$\boxed{\frac{AP}{PB} = \frac{2}{7}}$$

$$\boxed{2:7}$$

#LP: Find the ratio in which the y-axis divides the line segment joining the points (5, -6) and (-1, -4). Also, find the coordinates of the point of division.



$$\text{Let } \frac{AP}{PB} = \frac{k}{1}$$

$$P \left(\frac{-k+5}{k+1}, \frac{-4k-6}{k+1} \right)$$

Comparing 'x' coordinate

$$\frac{-k+5}{k+1} = 0$$

$$\begin{aligned} -k+5 &= 0 \\ +k &= 5 \end{aligned}$$

$$k = \frac{5}{1}$$

$$\boxed{5:1}$$

Comparing y-coordinate

$$y = \frac{-4k - 6}{k + 1}$$

$$y = \frac{-4(5) - 6}{5 + 1}$$

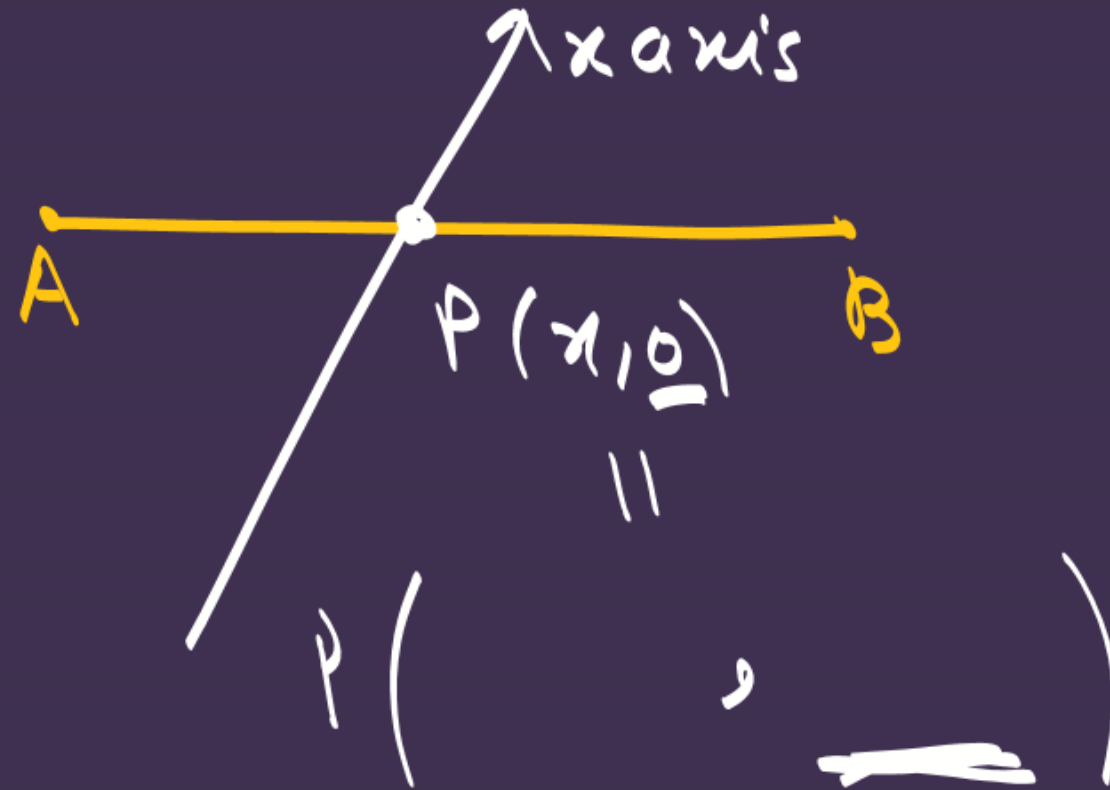
$$y = \frac{-26}{6}$$

$$y = -\frac{13}{3}$$

∴ Final Ans $P(0, -\frac{13}{3})$

#LP: In what ratio, does the x-axis divide the line segment joining the points $A(3, 6)$ and $B(-12, -3)$? (1 Mark) (2023)

- (a) 1 : 2
- (b) 1 : 4
- (c) 4 : 1
- (d) 2 : 1



#Mid - Point Formula:



$$M \left(\frac{x_2 + x_1}{1+1}, \frac{y_2 + y_1}{1+1} \right)$$

$$M \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$AM = MB$$

$$\frac{AM}{MB} = \frac{1}{1}$$

$$1:1$$

#LP: The mid-point of the line segment joining the points $P(-4, 5)$ and $Q(4, 6)$ lies on:

(a) x-axis

~~(b) y-axis~~

(c) origin

(d) neither x-axis nor y-axis

$$m\left(-\frac{-4+4}{2}, \frac{5+6}{2}\right)$$

$$m\left(\underline{0}, \frac{11}{2}\right)$$

#LP: If the mid-point of the line segment joining the points $(a, 4)$ and $(2, 2b)$ is $(2, 6)$, then the value of $(a + b)$ is given by:

- (a) 6
- (b) 7
- (c) 8
- (d) 16

given $\rightarrow$ $M(2, 6)$

by formula: $M\left(\frac{a+2}{2}, \frac{4+2b}{2}\right)$

Comparing

$\frac{a+2}{2} = 2$ $a+2 = 4$ $a = 2$	$\frac{4+2b}{2} = 6$ $4+2b = 12$ $2b = 8$ $b = 4$
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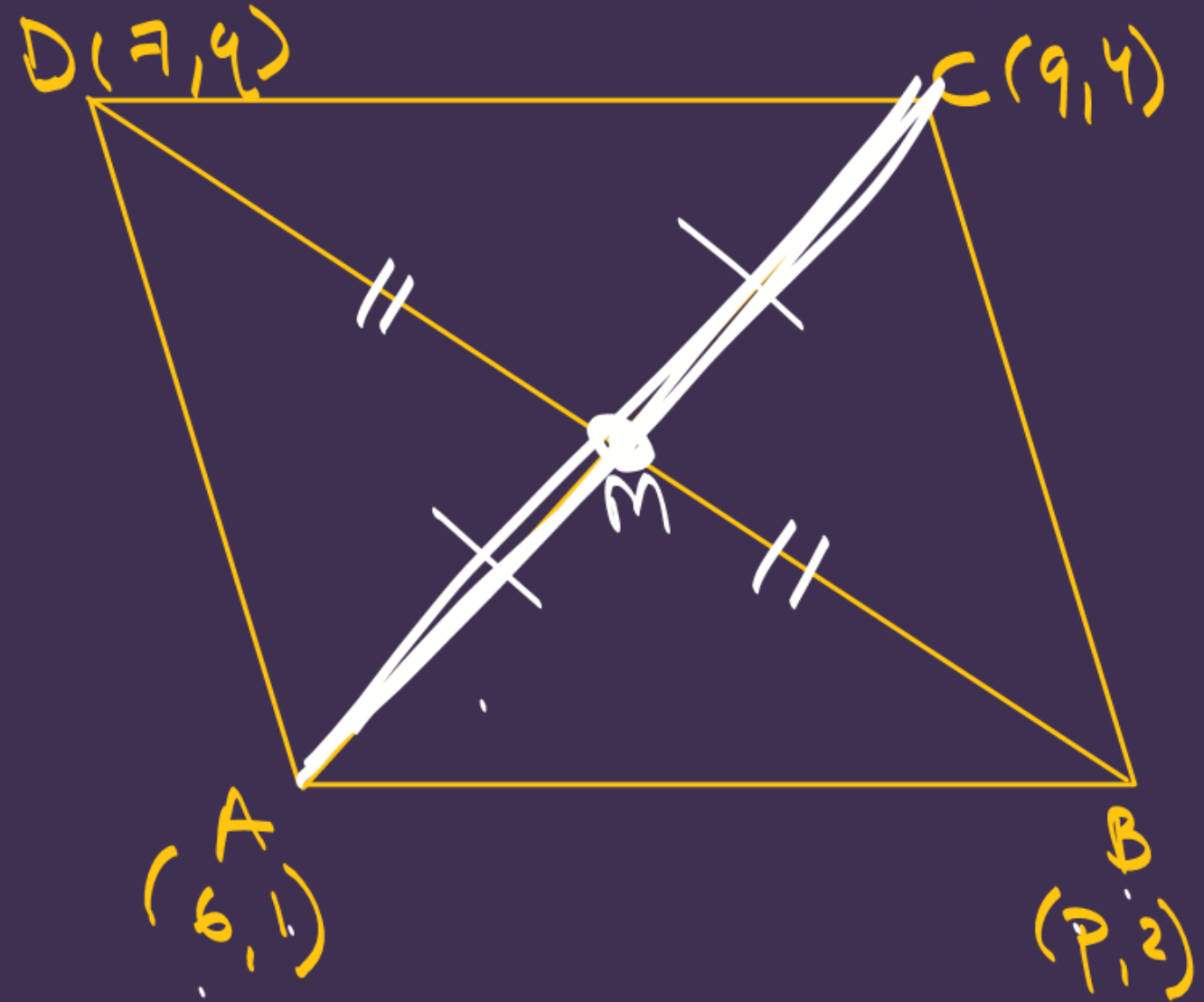
#LP: If the points $A(6, 1)$, $B(p, 2)$, $C(9, 4)$ and $D(7, q)$ are the vertices of a parallelogram $ABCD$, then find the values of p and q . H

$ABCD \rightarrow$ llgm $\rightarrow$ diagonals of llgm bisect each other

$$M_{AC} = M_{BD}$$
$$\left(\frac{6+9}{2}, \frac{1+4}{2} \right) = \left(\frac{p+7}{2}, \frac{q+2}{2} \right)$$

comparing $\frac{15}{2} = \frac{p+7}{2}$ $\boxed{p=8}$ ✓

comparing $5 = \frac{q+2}{2}$ $\boxed{q=8}$ ✓



Find the coordinates of the point C which lies on the line AB produced such that $AC = 2BC$, where coordinates of points A and B are $(-1, 7)$ and $(4, -3)$ respectively.

[CBSE 2025]

2



by formula $\Rightarrow B \left(\frac{x-1}{2}, \frac{y+7}{2} \right)$

$$\frac{x-1}{2} = 4$$

$$x-1 = 8$$

$$\boxed{x = 9}$$

$$\frac{y+7}{2} = -3$$

$$y+7 = -6$$

$$\boxed{y = -13}$$

$$\boxed{C(9, -13)}$$

#LP: If the mid-point of the line segment joining the points A(3, 4) and B(k, 6) is P(x, y) and $x + y - 10 = 0$, find the value of k.

formula $\rightarrow$ $P\left(\frac{3+k}{2}, \frac{4+6}{2}\right)$

$\frac{3+k}{2} = x$

$y = \frac{4+6}{2} = 5$

$$x + y - 10 = 0$$

$$\rightarrow \left(\frac{3+k}{2}\right) + (5) - 10 = 0$$

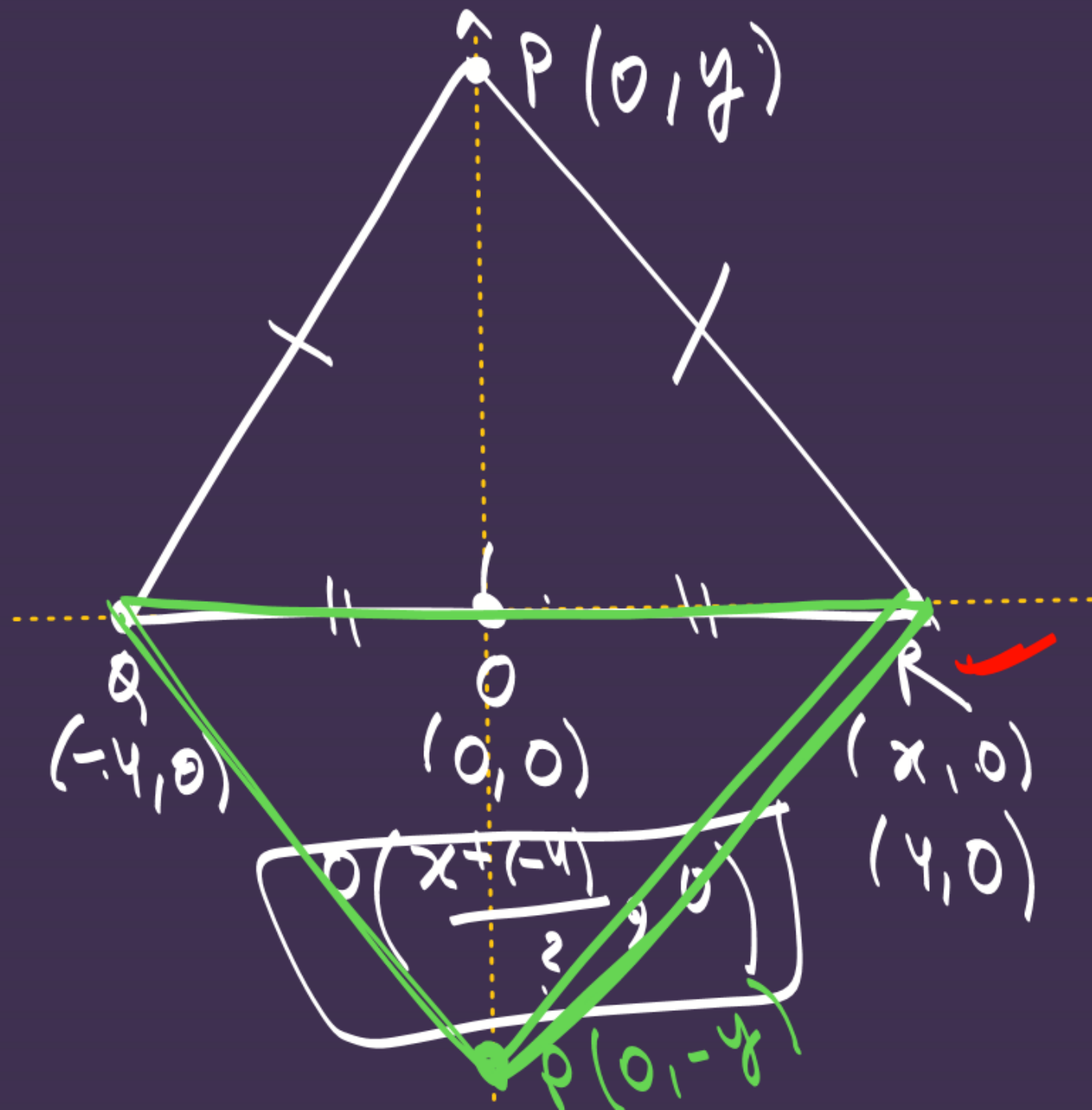
$$\rightarrow \frac{3+k}{2} - 5 = 0$$

$$\rightarrow \frac{3+k}{2} = 5$$

$$\rightarrow 3+k = 10$$

$$\boxed{k = 7}$$

#LP: The base QR of an equilateral triangle PQR lies on X-axis. The co-ordinates of the point Q are $(-4, 0)$ and origin is the mid-point of the base. Find the co-ordinates of the points P and R. **[CBSE 2015]**



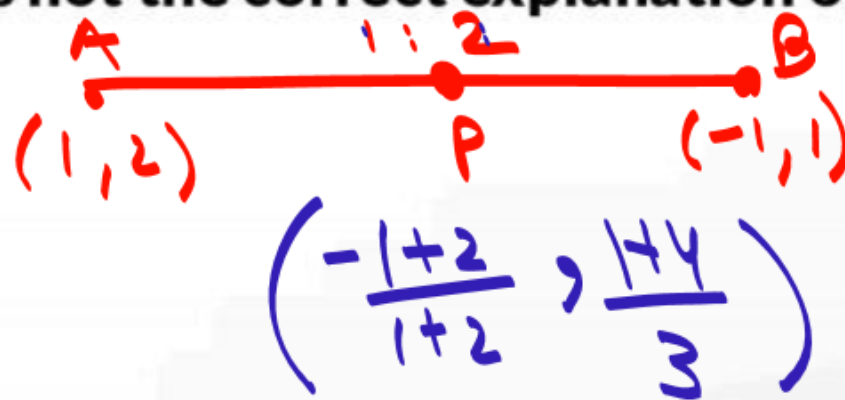
comparing, $\frac{x-4}{2} = 0$
 $x-4=0$
 $x=4$

$PQ = QR$
 $PQ^2 = QR^2$
 $(-4-0)^2 + (y-0)^2 = (8)^2$
 $16 + y^2 = 64$
 $y^2 = 64 - 16$
 $y^2 = 48$
 $y = \pm \sqrt{48}$
 $P(0, \pm \sqrt{48})$

#LP: Assertion (A): The point which divides the line segment joining the points A (1, 2) and B (-1, 1) internally in the ratio 1 : 2 is $(\frac{-1}{3}, \frac{5}{3})$. **(F)**

Reason (R): The coordinates of the point which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m_1 : m_2$ are $(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2})$ (1 Mark) (CBSE 2024)

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- ☒ (d) Assertion (A) is false but reason (R) is true.



$$P(\frac{1}{3}, \frac{5}{3})$$

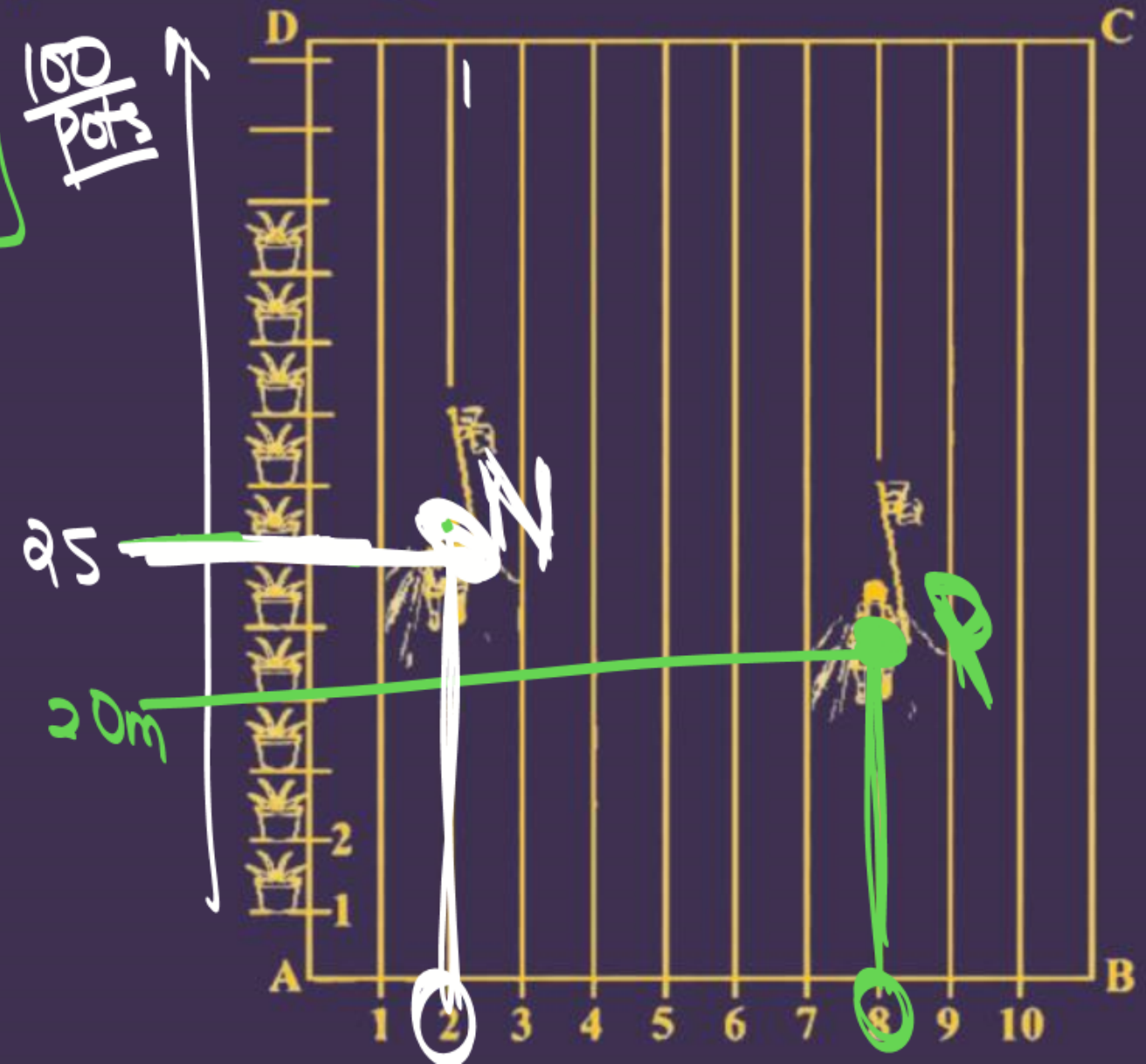
#LP: To conduct Sports Day activities, in your rectangular shaped school ground ABCD, lines have been drawn with chalk powder at a distance of 1m each. 100 flower pots have been placed at a distance of 1m from each other along AD as shown in Fig.. Niharika runs $\frac{1}{4}$ th the distance AD on the 2nd line and posts a green flag. Preet runs $\frac{1}{5}$ th the distance AD on the eighth line and posts a red flag. What is the distance between both the flags? If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, where should she post her flag?

(N) $\frac{1}{4} AD \Rightarrow \frac{1}{4} \times 100 \Rightarrow 25m$ $N(2, 25)$ $\frac{100 \text{ pots}}{1}$

(P) $\frac{1}{5} AD \Rightarrow \frac{1}{5} \times 100 \Rightarrow 20m$ $P(8, 20)$

i) $NP = \sqrt{(8-2)^2 + (25-20)^2}$

ii)



$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

$N(2, 25)$



M



P

$(8, 20)$

Mid-point

$$R\left(\frac{8+2}{2}, \frac{20+25}{2}\right)$$

$$\boxed{R\left(5, \frac{45}{2}\right)}$$

13/14 Completed!

Thank you Goodies !!

